

1. Express each sum as single vector.

a) $\vec{CA} + \vec{TC} - \vec{RC} + \vec{RT}$

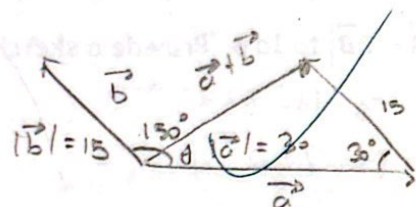
$$\begin{aligned} &= \vec{CA} + \vec{TC} + \vec{CR} + \vec{RT} \\ &= \vec{CR} + \vec{CT} + \vec{TC} + \vec{CA} \\ &= \vec{CA} \end{aligned}$$

b) $\vec{AB} - \vec{CB} - \vec{AD} + \vec{CD}$

$$\begin{aligned} &= \vec{AB} + \vec{BC} + \vec{DA} + \vec{CD} \\ &= \vec{AB} + \vec{BC} + \vec{CD} + \vec{DA} \\ &= \vec{0} \end{aligned}$$

2. Two vectors, $\vec{a}$ and $\vec{b}$, have an angle of 150° between them. The vectors are such that $|\vec{a}| = 30$ and $|\vec{b}| = 15$. Determine to one d.p.

a) $|\vec{a} + \vec{b}|$



$$\begin{aligned} |\vec{a} + \vec{b}| &= \sqrt{30^2 + 15^2 - 2(30)(15)\cos 30^\circ} \\ &= \sqrt{1125 - 900\cos 30^\circ} \end{aligned}$$

$$|\vec{a} + \vec{b}| = 18.6$$

b) the angle between $\vec{a}$ and $\vec{a} + \vec{b}$.

$$\frac{\sin \theta}{15} = \frac{\sin 30^\circ}{18.6}$$

$$\sin \theta = \frac{15 \sin 30^\circ}{18.6}$$

$$\theta = \sin^{-1}\left(\frac{15 \sin 30^\circ}{18.6}\right)$$

$$\theta = 23.8^\circ$$

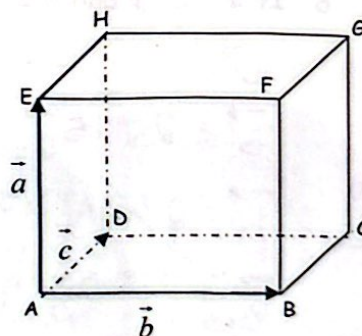
3. A cube is constructed from the three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, as shown below. Express each diagonals in terms of $\vec{a}$, $\vec{b}$ and $\vec{c}$.

a) $\vec{CE} - \vec{GB}$

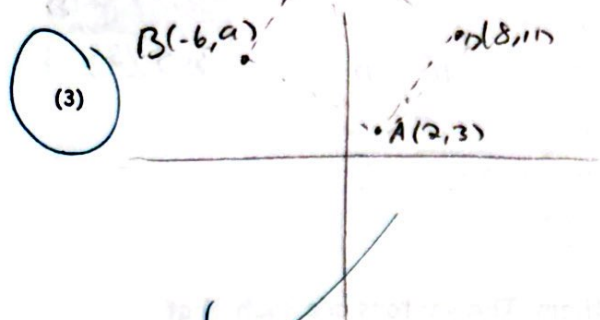
$$\begin{aligned} &= (\vec{a} - \vec{b} - \vec{c}) - (-\vec{a} - \vec{c}) \\ &= \vec{a} - \vec{b} - \vec{c} + \vec{a} + \vec{c} \\ &= 2\vec{a} - \vec{b} \end{aligned}$$

b) $\vec{BH} - \vec{AG}$

$$\begin{aligned} &= (\vec{a} - \vec{b} + \vec{c}) - (\vec{a} + \vec{b} + \vec{c}) \\ &= \vec{a} - \vec{b} + \vec{c} - \vec{a} - \vec{b} - \vec{c} \\ &= -2\vec{b} \end{aligned}$$



4. Rectangle ABCD has vertices at A(2,3), B(-6,9), C(x,y), and D(8,11). Determine point C.



Ans:-

$$\vec{CB} = (8 - 6, 11 - 9) = (2, 2)$$

$$\vec{BA} = (2, 3) - (-6, 9) = (8, -6)$$

$$\begin{aligned}\vec{BC} &= \vec{AD} \\ \vec{AD} &= \vec{OD} - \vec{OA} \\ &= (8, 11) - (2, 3)\end{aligned}$$

$$= (6, 8)$$

$$\begin{aligned}\vec{BC} &= \vec{OC} - \vec{OB} \\ &= (x, y) - (-6, 9) \\ &= (x+6, y-9)\end{aligned}$$

$$(6, 8) = (x+6, y-9)$$

$$x+6 = 6$$

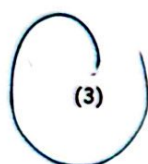
$$x = 0$$

$$y-9 = 8$$

$$y = 17$$

$$C(0, 17)$$

5. Given, $|\vec{A}| = 4$ and $|\vec{B}| = 5$ and $|\vec{A} + \vec{B}| = 6$, determine $|\vec{A} - 2\vec{B}|$ to 1d.p. Provide a sketch.

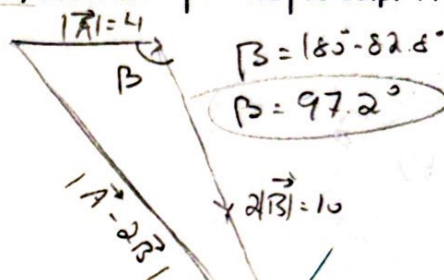


$$6^2 = 4^2 + 5^2 - 2(4)(5)\cos\theta$$

$$-5 = -40\cos\theta$$

$$\cos\theta = \frac{1}{8}$$

$$\theta = 82.8^\circ$$



$$\theta = 180^\circ - 82.8^\circ$$

$$\theta = 97.2^\circ$$

$$|\vec{A} - 2\vec{B}| = \sqrt{4^2 + 10^2 - 2(4)(10)\cos 97.2^\circ}$$

$$= \sqrt{116 - 80\cos 97.2^\circ}$$

$$|\vec{A} - 2\vec{B}| = 11.2$$

6. If $\vec{a} = \vec{x} - \vec{y}$ and $\vec{b} = 2\vec{x} + 5\vec{y}$, express $\vec{x}$ and $\vec{y}$ in terms of $\vec{a}$ and $\vec{b}$

$$\vec{a} = \vec{x} - \vec{y} \longrightarrow \vec{x} = \vec{a} + \vec{y}$$

$$\vec{b} = 2\vec{x} + 5\vec{y}$$

$$\vec{b} = 2(\vec{a} + \vec{y}) + 5\vec{y}$$

$$\vec{b} = 2\vec{a} + 2\vec{y} + 5\vec{y}$$

$$\vec{b} - 2\vec{a} = 7\vec{y}$$

$$\vec{y} = \frac{\vec{b} - 2\vec{a}}{7}$$

$$\vec{x} = \vec{a} + \left(\frac{\vec{b} - 2\vec{a}}{7}\right)$$

$$\vec{x} = \frac{7\vec{a} + \vec{b} - 2\vec{a}}{7}$$

$$\vec{x} = \frac{5\vec{a} + \vec{b}}{7}$$

7. Determine an equation to describe the set of points equidistant from $A(-2, -6, 3)$ and $B(3, -4, 12)$

Let P be point that is equidistant from A and B

$$P(a, b, c)$$

$$|AP| = |BP|$$

$$\vec{AP} = \vec{OP} - \vec{OA}$$

$$\vec{AP} = (a, b, c) - (-2, -6, 3)$$

$$\vec{AP} = (a+2, b+6, c-3)$$

$$\vec{BP} = \vec{OP} - \vec{OB}$$

$$= (a, b, c) - (3, -4, 12)$$

$$\vec{BP} = (a-3, b+4, c-12)$$

$$|\vec{AP}| = \sqrt{(a+2)^2 + (b+6)^2 + (c-3)^2}$$

$$|\vec{BP}| = \sqrt{(a-3)^2 + (b+4)^2 + (c-12)^2}$$

$$(a+2)^2 + (b+6)^2 + (c-3)^2 = (a-3)^2 + (b+4)^2 + (c-12)^2$$

$$4a + 12b - 6c + 49 = -6a + 8b - 24c + 169$$

$$10a + 4b + 18c - 120 = 0$$

8. If $\vec{a} = (3, 2, 1)$, $\vec{b} = (5\vec{i} + 5\vec{k})$, $\vec{c} = 4\vec{j}$, calculate $|\vec{a} - 2\vec{b} + \frac{1}{2}\vec{c}|$

$$\vec{a} = 3\vec{i} + 2\vec{j} + \vec{k}$$

$$\vec{b} = 5\vec{i} + 5\vec{k}$$

$$\vec{c} = 4\vec{j}$$

$$\vec{a} - 2\vec{b} + \frac{1}{2}\vec{c}$$

$$= (3\vec{i} + 2\vec{j} + \vec{k}) - 2(5\vec{i} + 5\vec{k}) + \frac{1}{2}(4\vec{j})$$

$$= 3\vec{i} + 2\vec{j} + \vec{k} - 10\vec{i} - 10\vec{k} + 2\vec{j}$$

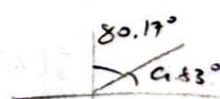
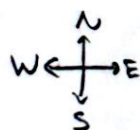
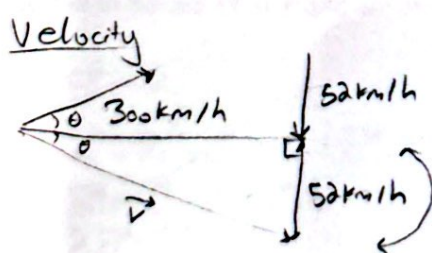
$$= -7\vec{i} + 4\vec{j} - 9\vec{k}$$

$$\vec{a} - 2\vec{b} + \frac{1}{2}\vec{c} = (-7, 4, -9)$$

$$|\vec{a} - 2\vec{b} + \frac{1}{2}\vec{c}| = \sqrt{(-7)^2 + (4)^2 + (-9)^2}$$

$$= \sqrt{146}$$

9. Mr. DaSilva is so excited to fly his first plane. He wants to visit his buddy who is exactly 850km east of him. There is a strong 52km/h wind coming directly from the north. If he's flying at 300km/h, at what bearing must he adjust his flight to end up directly at his buddies place? And how long would it take him to get there?



$$\frac{850}{304.47} = 2.79 \text{ hrs}$$

Therefore, Mr. DaSilva must fly at a bearing of 80.17° instead of due east to account for the wind. It will take him 2.79 hrs to get there.

$$|\vec{V}| = \sqrt{300^2 + 52^2}$$

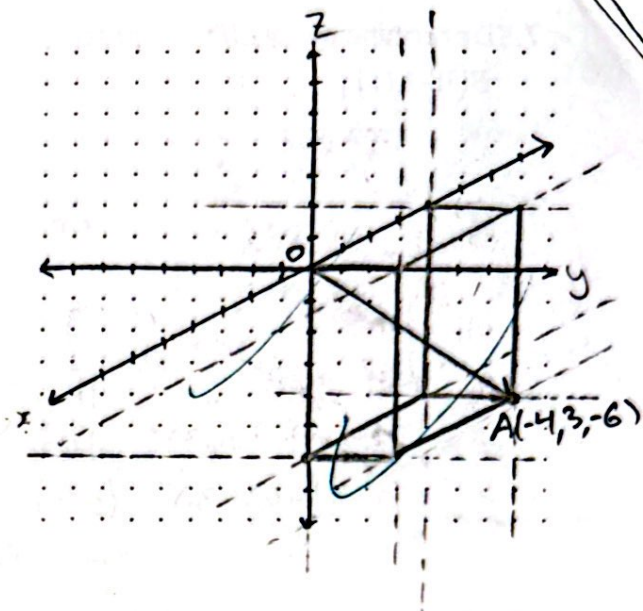
$$|\vec{V}| = 304.47 \text{ km/h}$$

$$\tan \theta = \frac{52}{300}$$

$$\theta = \tan^{-1}\left(\frac{52}{300}\right)$$

$$\theta = 9.83^\circ$$

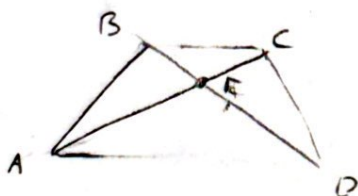
10. Plot the point $A(-4, 3, -6)$ in $\mathbb{R}^3$, using a rectangular prism. Draw the vector $\overrightarrow{OA}$.



(2)

10. ABCD is a trapezoid whose diagonals AC and BD intersect at the point E.

If $\frac{2}{5}\overrightarrow{DE} = \overrightarrow{EB}$, prove that $\frac{7}{3}\overrightarrow{AE} = \frac{5}{3}\overrightarrow{AB} + \frac{2}{3}\overrightarrow{AD}$.



(4)

$$\frac{2}{5}\overrightarrow{DE} = \overrightarrow{EB} \rightarrow -\frac{2}{5}\overrightarrow{DE} = -\overrightarrow{EB} \rightarrow \frac{2}{5}\overrightarrow{ED} = \overrightarrow{BE}$$

$$\frac{7}{3}\overrightarrow{AE} = \frac{5}{3}\overrightarrow{AB} + \frac{2}{3}\overrightarrow{AD}$$

$$\overrightarrow{AB} = \overrightarrow{AE} + \overrightarrow{EB}$$

$$\overrightarrow{AD} = \overrightarrow{AE} + \overrightarrow{ED}$$

$$\frac{7}{3}\overrightarrow{AE} = \frac{5}{3}(\overrightarrow{AE} + \overrightarrow{EB}) + \frac{2}{3}(\overrightarrow{AE} + \overrightarrow{ED})$$

$$\frac{7}{3}\overrightarrow{AE} = \frac{5}{3}\overrightarrow{AE} + \frac{5}{3}\left(\frac{2}{5}\overrightarrow{DE}\right) + \frac{2}{3}\overrightarrow{AE} + \frac{2}{3}\overrightarrow{ED}$$

$$\frac{7}{3}\overrightarrow{AE} = \frac{7}{3}\overrightarrow{AE} + \frac{2}{3}\overrightarrow{DE} - \frac{2}{3}\overrightarrow{DE}$$

$$\frac{7}{3}\overrightarrow{AE} = \frac{7}{3}\overrightarrow{AE}$$

$$LS = RS$$